Excel Data Exploration – Data Analytics – Module 1

Case Study - Product Database		
Basic Data Exploration		
Case Questions	**Findings**	**Formula Used**
1. Sum, Count, and Average **Perform the following calculations using Excel functions:**		
• Calculate the total price of all products in the dataset.	10100	SUM(Product_DB[Price ($)])
• Determine how many products exist in the dataset.	21	Remove Duplicates
	34	COUNT(Product_DB[Price ($)])
• Calculate the average price of the products.	297.0588235	AVERAGE(Product_DB[Price ($)])
2. Minimum and Maximum Values **Using Excel functions:**		
• Identify the minimum price among all products	30	MIN(Product_DB[Price ($)])
• Identify the maximum price among all products.	1000	MAX(Product_DB[Price ($)])
3. Logical Function – IF **• Create a new column named Price Range.**		
• Use the IF function to categorize products based on their price:		
○ If Price ≥ $500 → classify as High Price		IF([@[Price ($)]]>=500,"High Price","Standard Price")
○ If Price < $500 → classify as Standard Price		
4. Conditional Functions – SUMIF and COUNTIF **Perform the following conditional calculations:**		
• Use the SUMIF function to calculate the total price of products in the Electronics ca	8050	SUMIF(Product_DB[Category],"Electronics",Product_DB[Price ($)])
• Use the COUNTIF function to determine the number of products with a price less th	11	COUNTIF(Product_DB[Price ($)],"<100")
5. Text Formatting – LEFT, RIGHT, MID • Use text functions to extract information from the Product ID column. • Create the following new columns:		
○ Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.		LEFT([@[Product ID]],2)
○ Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.		RIGHT([@[Product ID]],2)
○ Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.		MID([@[Product ID]],4,3)

3.Logical Functions: Using IF

Product ID	Product Nan	Brand Nam	Price ($)	Quantity	Category	Price Rang
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price
16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price
07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price
02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price
09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price

4.Text Formating : Using LEFT,MID,RIGHT

Product ID	Product Nan	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Month	Country Cod
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	JAN	US
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	FEB	US
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	MAR	US
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	APR	US
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	MAY	US
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	JUN	UK
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	JUL	UK
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	AUG	UK
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	SEP	UK
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	OCT	UK
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	JUN	IN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	NOV	AU
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DEC	DE
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	FEB	CA
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	APR	ES
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	AUG	CA
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	AUG	CN
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	JAN	IT
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	MAR	UK
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	AUG	US
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	MAY	RU
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	JAN	CA
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	JUL	BR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	AUG	CA
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	SEP	CA
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16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	APR	ES
07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	MAR	CA
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	APR	CA
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	MAY	CA
02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	DEC	CA
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	JUN	IN
09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price	09	JUL	FR